

Zadanie 13 [3pkt] - TYP 3.3 (metoda 1)

Rozwiąż równanie:  $(2x + 5)(x + 1)(x - 5) = 5x - 25$

$$(2x+5)(x+1)(x-5) = 5x-25$$

$$(2x+5)(x+1)(x-5) = 5(x-5) \quad | -5(x-5)$$

$$(2x+5)(x+1)(x-5) - 5(x-5) = 0$$

$$(x-5)[(2x+5)(x+1) - 5] = 0$$

$$(x-5)(2x^2+2x+5x+5-5) = 0$$

$$(x-5)(2x^2+7x) = 0$$

$$(x-5) \cdot x \cdot (2x+7) = 0$$

$$\begin{aligned} x-5 &= 0 \\ x_1 &= 5 \end{aligned}$$

$$x_2 = 0$$

$$\begin{aligned} 2x+7 &= 0 \\ 2x &= -7 \\ x_3 &= -\frac{7}{2} \end{aligned}$$

Odp:  $x \in \{-\frac{7}{2}, 0, 5\}$ .